

CE 310 — Week 13 Weekly Quiz

Week: 13 | **Topic:** Monte Carlo Simulation

Due: Closes 8:00 AM Tuesday, Nov 24 — the day before Exercise 13 and Homework 13 are due. Taken on D2L

Points: 15 questions × 1 pt each = 15 pts

Attempts: Two — the higher score counts

Time limit: 25 minutes

Instructions: Answer each question in the format it asks for — most are multiple choice, with one matching, one ordering, one select-all-that-apply and one true/false. This quiz is auto-graded and must be submitted before it closes at the time given above. You may use your notes and readings, but work independently.

This quiz covers the Monday lecture and the Wednesday Python tutorial — the purpose of Monte Carlo simulation, random sampling, analytical versus simulated uncertainty, percentile interpretation, and variance decomposition. The tutorial's warning boxes are where the most time-consuming mistakes are flagged, so they are worth reading before you start.

Questions

Q1. Monte Carlo simulation differs from one-at-a-time (OAT) sensitivity analysis primarily because Monte Carlo:

- A) Requires fitting a new regression model for each scenario
- B) Only works when all inputs are normally distributed
- C) Replaces the regression model with a lookup table of pre-computed values
- D) Varies all uncertain inputs at once according to probability distributions

Q2. [True/False] Seeding the generator — whether with `np.random.seed(42)` or `np.random.default_rng(42)` — makes a simulation reproducible without making its draws any less random.

Q3. The item cost model is $\text{cost} = Q \times P \times (1 + \text{OH}) \times (1 + \text{Esc})$, with a base cost of **\$50,459**. The take-off quantity is drawn from $\text{Normal}(2000, 182)$ and the other three inputs are held at their base values. What is $E[\text{cost}]$?

- A) $\$50,459 + 182 \times 21.26 \times 1.13 \times 1.05 = \$55,060$, adding one standard deviation
- B) $\$50,459$ — because $E[Q] = 2,000$, and cost is linear in Q
- C) $\$4,601$, the standard deviation of the simulated costs
- D) It cannot be determined without running the simulation

Q4. Same quantity-only setup, with $\sigma_Q = 182$ CY, $P = \$21.26/\text{CY}$, overhead 13% and escalation 5%. What is the **analytical** standard deviation of item cost?

- A) 182 CY — the same as the input standard deviation, since cost tracks quantity
- B) $\$21.26 \times 1.13 \times 1.05 = \25.23 , leaving out the σ_Q multiplier entirely
- C) $\$50,459 / 182 = \277 , dividing the base cost by the input's spread
- D) $182 \times 21.26 \times 1.13 \times 1.05 \approx \$4,601$

Q5. An estimator wants to set a contingency so there is a 90% probability that the actual item cost lands at or below the bid. Which percentile of the Monte Carlo cost distribution should be used?

- A) 10th percentile (P10)
 - B) 90th percentile (P90)
 - C) 50th percentile (median)
 - D) 95th percentile (P95)
-

Q6. [Select all that apply] A variance decomposition of item cost attributes **59%** of the variance to unit price and **36%** to take-off quantity. Which of the following follow?

- (A) Narrowing the price range will cut cost variance more than an equally proportional narrowing of the quantity range
 - (B) Unit price is the larger of the two contributors to cost uncertainty
 - (C) Removing quantity uncertainty entirely would eliminate roughly 36% of the cost variance
 - (D) Unit price has the larger effect on the mean item cost
 - (E) Unit price matters about 1.6 times as much as quantity, since $59 / 36 \approx 1.6$
-

Q7. A Monte Carlo produces 10,000 simulated item costs, and $P(\text{cost} > \$60,000) = 0.12$. Which interpretation is correct?

- A) In about 12% of the simulated jobs the item cost came in above \$60,000
 - B) The most this item could possibly cost is \$60,000, which 12% of jobs reach
 - C) The mean item cost is $\$60,000 \times 0.12 = \$7,200$ across the simulated jobs
 - D) There is an 88% probability that the item costs exactly \$60,000 to build
-

Q8. Run with three quotes in hand, the simulation gives a cost standard deviation of **\$7,716**. Re-run with the price drawn from the model's full residual spread instead — the whole market — and it becomes **\$36,594**. What does that tell you?

- A) The second run is wrong, because a standard deviation cannot rise by a factor of five
 - B) The mean cost is far lower in the first run than in the second
 - C) Almost all of the answer's precision comes from having quotes, not from the model
 - D) The two runs used different seeds, so they are not comparable
-

Q9. [Ordering] Put the steps of a Monte Carlo simulation in the order you would carry them out.

- (A) Assign a probability distribution to each uncertain input
 - (B) Draw one value from each distribution and evaluate the model once
 - (C) Repeat the draw-and-evaluate step thousands of times, collecting every output
 - (D) Read percentiles and exceedance probabilities off the collected distribution
-

Q10. [Matching] Match each simulation output to what it reports.

Choices:

- (A) The mean of the simulated outcomes
- (B) P10
- (C) P90
- (D) The P10–P90 range
- (E) $P(\text{cost} > \$60,000)$

Matches:

1. An 80% interval describing the spread of simulated outcomes
2. The value that 10% of simulated outcomes fall below

3. The fraction of simulated years exceeding a stated threshold
 4. The value that 90% of simulated outcomes fall below — the usual basis for a budget
-

Q11. Your Monte Carlo results differ slightly from a classmate's, although the code matches. What makes a simulation reproducible?

- A) Running more iterations, until the results stop moving between one run and the next
 - B) Sorting the samples before taking percentiles, so the ordering cannot vary
 - C) Seeding the generator — `np.random.default_rng(42)` — so the same draws come out every run
 - D) Using the same version of numpy and the same operating system as your classmate
-

Q12. You have 10,000 simulated annual costs in `sims`. How do you get P90?

- A) `np.percentile(sims, 0.90)` — the argument is a probability
 - B) `sims.mean() * 1.9` — scaling the mean by the percentile
 - C) `np.max(sims) * 0.90` — nine tenths of the largest simulated value
 - D) `np.percentile(sims, 90)` — the argument is a percentage, 0 to 100
-

Q13. How do you estimate P(annual cost exceeds \$500k) from the simulated array?

- A) `np.mean(sims > 500_000)`
 - B) `np.sum(sims > 500_000)`
 - C) `np.max(sims) - 500_000`
 - D) `np.percentile(sims, 500_000)`
-

Q14. Raising the simulation from 1,000 to 100,000 runs. What does that improve?

- A) It narrows the spread of the underlying cost distribution being simulated
 - B) It narrows the uncertainty in your ESTIMATE of that distribution — P90 stops jumping between runs
 - C) It makes the simulated output distribution closer to Normal in shape
 - D) It reduces the standard deviation of each of the uncertain inputs you specified
-

Q15. Variance decomposition attributes output variance to each input. What is it used for?

- A) Confirming that the uncertain inputs are independent of one another
- B) Ranking which input's uncertainty drives the most output uncertainty
- C) Checking that the simulation has converged on a stable answer
- D) Testing whether the simulated output distribution is Normal